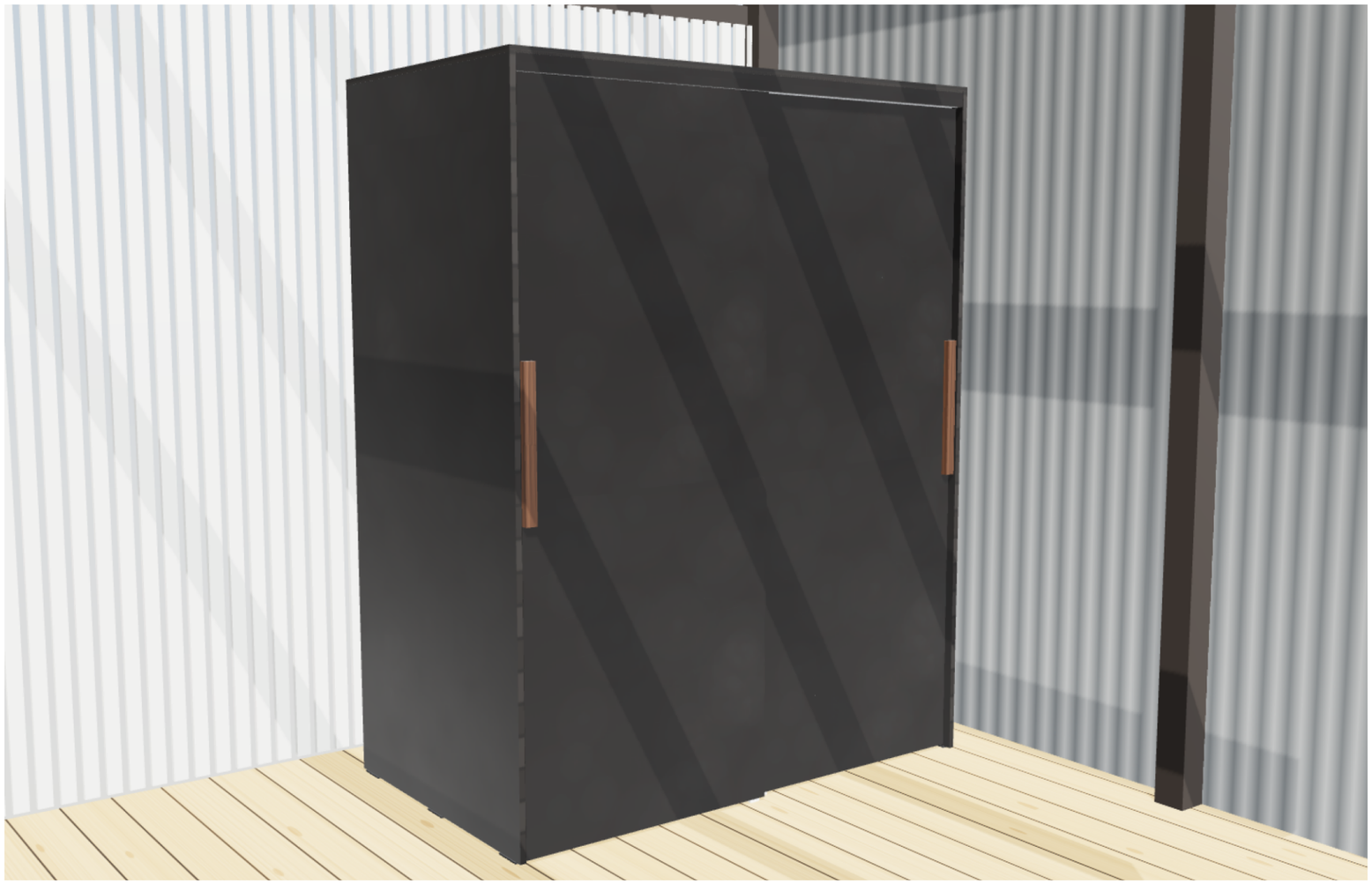
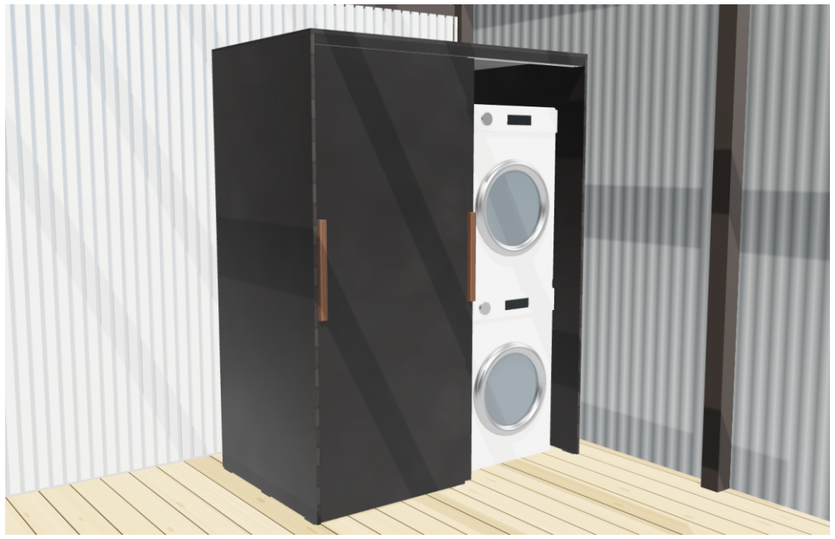
Outdoor appliance cabinet: fridge + stacked washer/dryer





A plain black formply box: flush top, the fridge door flush with the sides and a slim fascia (3 mm shadow line), Brio top-hung door gear, no front vents, and two short American walnut pulls.

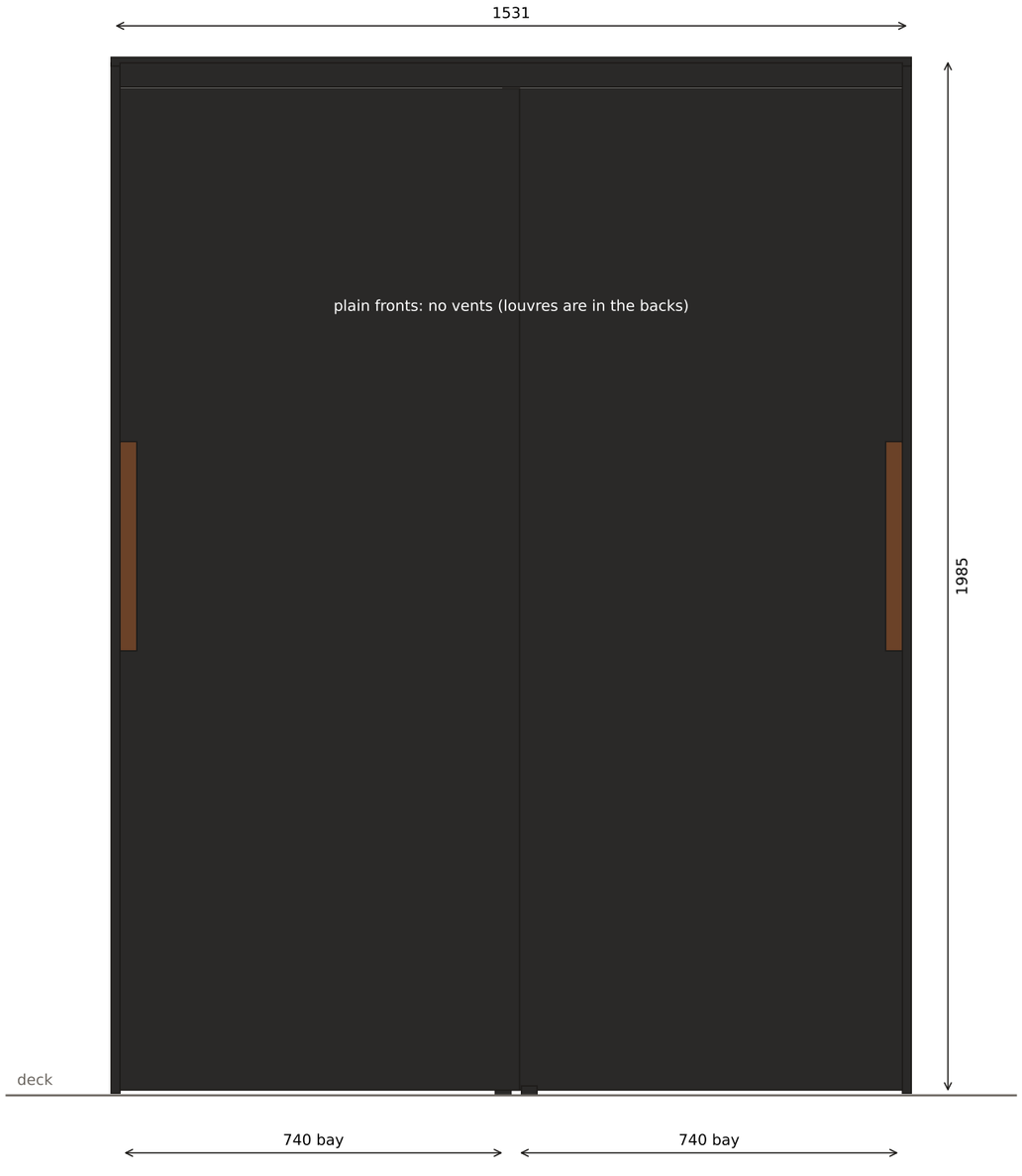
1531 W × 900 D × 1985 H. Two 740 clear bays. 8 CNC parts on 8 sheets of 2400 × 1200 × 17.

The fridge breathes through rain-hooded openings low and high in the hidden back panels (about 290 cm2 net each level). Run the dryer with the laundry door slid open. The box sits on 4 mm ASA pads, held down by two stainless angles into the deck joists.

18 of 18 design checks pass (sheet 6). Open items must be closed before the cut.

General arrangement

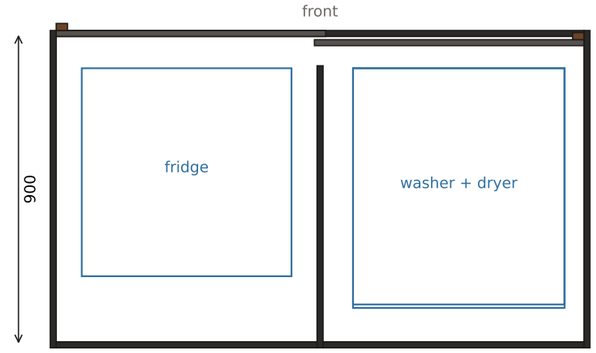
Front elevation (doors closed)



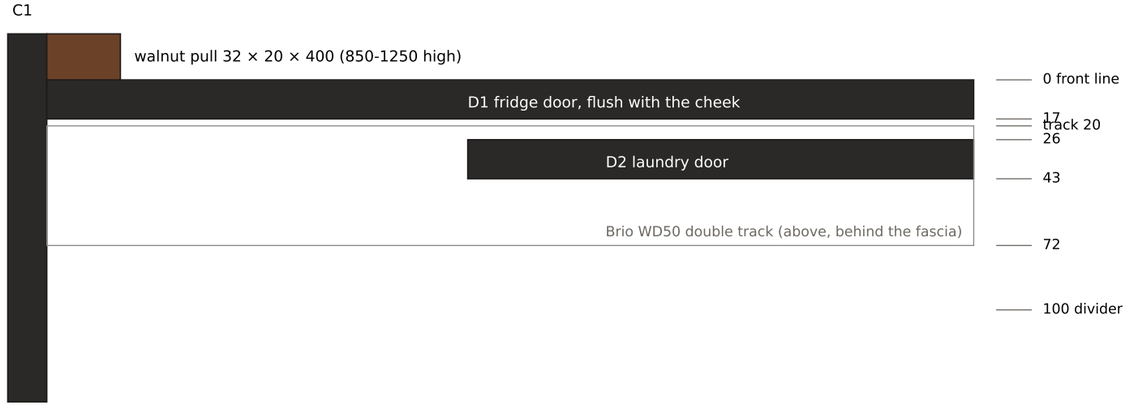
Side elevation (studio cheek, outside)



Plan (top removed)

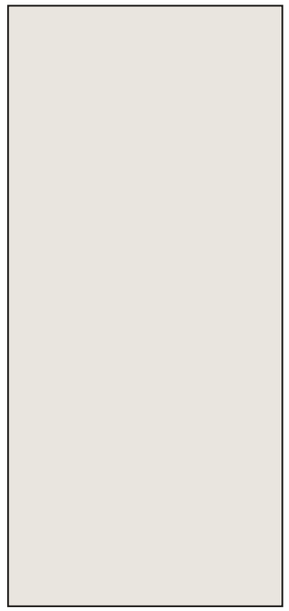


Section through the doors and tracks (plan, enlarged)



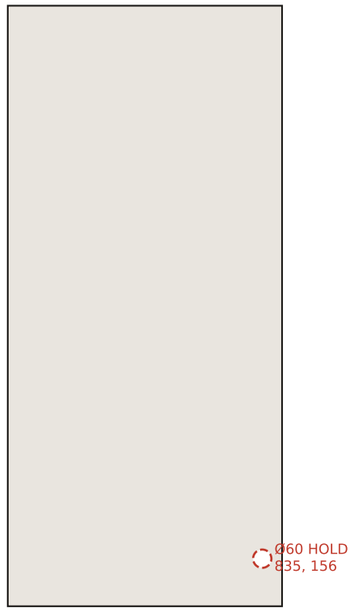
CNC part drawings

C1 Left cheek 900 × 1970 ×1



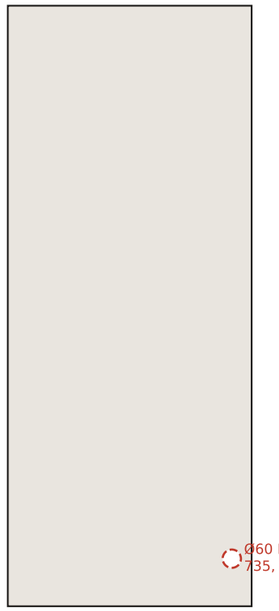
X = 0 at the cabinet front edge, Y = 0 at the bottom edge.

C3 Studio cheek 900 × 1970 ×1



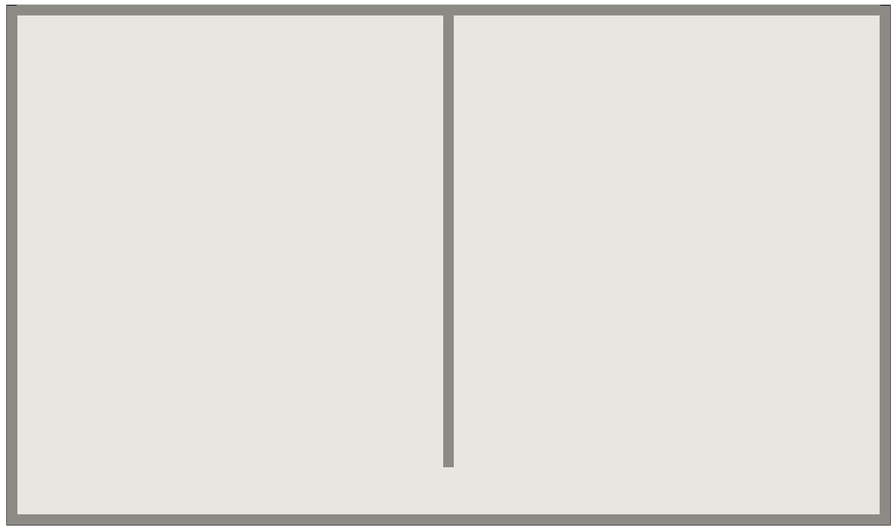
X = 0 at the cabinet front edge (mirror view: routed from the outside face). HOLD hole: confirm on site.

C2 Divider 800 × 1970 ×1



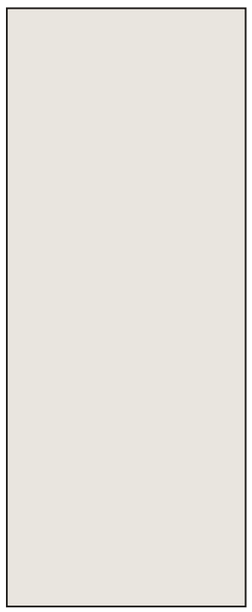
X = 0 at the front edge. HOLD hole: confirm on site.

C4 Top 1531 × 900 ×1



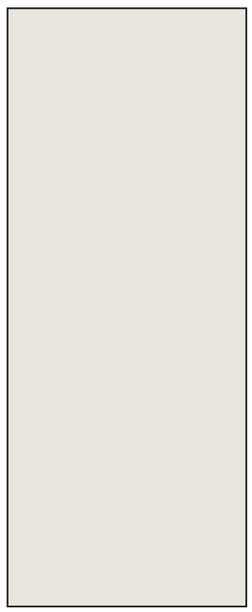
Routed on the UNDERSIDE: 6 mm housings for every vertical. X = 0 at the kitchen-side edge, Y = 0 at the front edge.

D1 Fridge door 764.5 × 1915 ×1



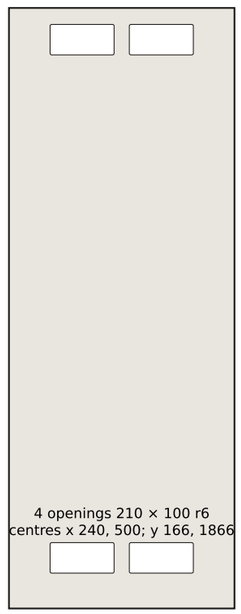
Plain panel. 5 x 20 sawcut groove along the bottom edge for the Brio 101N guide, cut in the workshop and sealed.

D2 Laundry door 764.5 × 1915 ×1



Plain panel. 5 x 20 sawcut groove along the bottom edge for the Brio 101N guide, cut in the workshop and sealed.

C5 Back, fridge bay 740 × 1970 ×1



4 openings 210 x 100, r6 corners, for the ASA louvres. Drawn from the inside face, X = 0 at the left.

C7 Fascia 1497 × 46 ×1

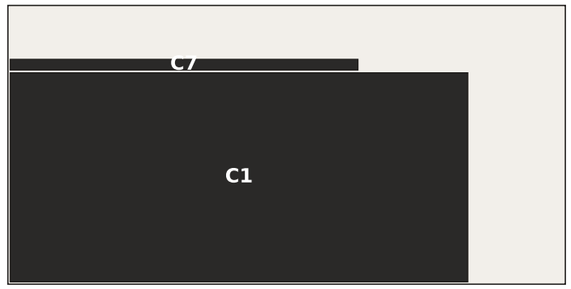


C6 Back, laundry bay: as C5.
C4 is routed on its UNDERSIDE: 6 deep housings (grey) for C1, C2, C3, the backs and the head rail.
Cheek housings are 0.6 over the panel for glue and fit.

Cut list and indicative nests

Code	Part	Size	Qty	Notes
C1	Left cheek	900 × 1970	1	X = 0 at the cabinet front edge, Y = 0 at the bottom edge.
C2	Divider	800 × 1970	1	X = 0 at the front edge. HOLD hole: confirm on site.
C3	Studio cheek	900 × 1970	1	X = 0 at the cabinet front edge (mirror view: routed from the outside face). HOLD hole: confirm on site.
C4	Top	1531 × 900	1	Routed on the UNDERSIDE: 6 mm housings for every vertical. X = 0 at the kitchen-side edge, Y = 0 at the [...]
C5	Back, fridge bay	740 × 1970	1	4 openings 210 x 100, r6 corners, for the ASA louvres. Drawn from the inside face, X = 0 at the left.
C6	Back, laundry bay	740 × 1970	1	As C5.
C7	Fascia	1497 × 46	1	
D1	Fridge door	764.5 × 1915	1	Plain panel. 5 x 20 sawcut groove along the bottom edge for the Brio 101N guide, cut in the workshop and [...]
D2	Laundry door	764.5 × 1915	1	Plain panel. 5 x 20 sawcut groove along the bottom edge for the Brio 101N guide, cut in the workshop and [...]

S1 2400 × 1200 × 17



S2 2400 × 1200 × 17



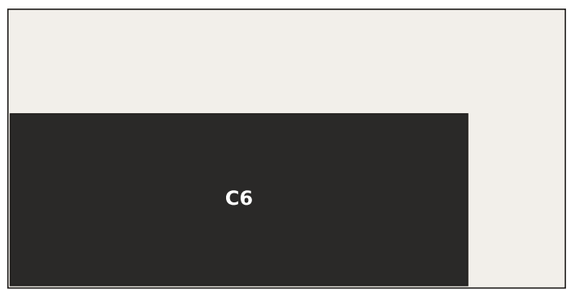
S3 2400 × 1200 × 17



S4 2400 × 1200 × 17



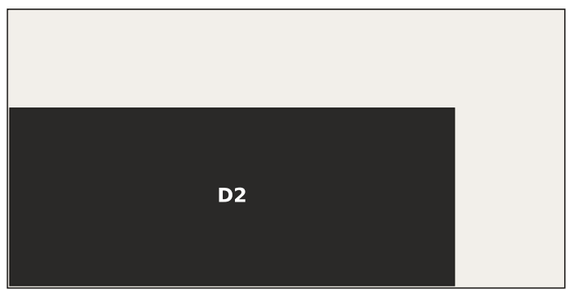
S5 2400 × 1200 × 17



S6 2400 × 1200 × 17



S7 2400 × 1200 × 17



S8 2400 × 1200 × 17



Build steps and shopping list

1. Confirm before cutting

Confirm the washer and dryer models and their stacking kit (dryer must be heat-pump or condenser). Buy the Brio WD50-15/2 kit and check its track against data sheet B55 before the fascia and door heights are released. Measure the power point position along the studio wall (it is about 100 above the deck) and the gaps to the slat screen and studio wall.

2. Seal, groove and print

Seal every CNC-cut edge with two coats of exterior edge sealer, bottom edges most of all. Cut the 5 x 20 sawcut groove along the bottom edge of each door for the Brio guide, and seal it. Print the ASA parts (Prints tab).

3. Pads on the deck

Mark the footprint. Bed the thirteen 4 mm ASA edge pads in sealant on the lines of the cheeks, divider and backs, one 10g x 25 each.

4. Carcass on the top, upside down

Lay the top film-face down on padded trestles, housings up. Run PU sealant in each housing. Stand C1, C2, C3, C5 and C6 in their housings; the housings square the box. Screw down through the top into each edge, 8g x 50 stainless at 200 centres, countersunk and sealed.

5. Angles and rain hoods

Fit the four black aluminium angles inside the back corners. Staple stainless mesh behind the eight back openings, then bed each ASA rain hood in sealant over its opening from the outside, mouth down, 4 x 6g x 16. Run a bead along the top edge of each hood.

6. Stand it up and hold it down

Two people. Turn the carcass upright onto the pads. Check plumb and diagonals. Screw the two stainless hold-down angles inside the cheeks into the deck joists and the cheeks.

7. Fascia and track

Push the fascia up into its housing between the cheeks, flush with their fronts, and screw down through the top. Cut the Brio double track to 1497 and screw it up into the top behind the fascia, set out to the Brio data sheet so the fridge door hangs flush with the cheeks.

8. Doors and pulls

Screw the walnut pulls on from behind: on the left edge of D1 and the right edge of D2, 850-1250 above the deck. Screw the Brio face-fixed hangers to the back face of each door, hang D1 on the front line and D2 on the rear, and adjust to 10 off the deck with a 3 mm line under the fascia. Set the Brio door stop so D1 parks 40 short of the studio cheek.

9. Floor guides

Screw the two Brio 101N guides to the deck where each door always passes over them (about 750 and 800 from the kitchen side), engaged in the door grooves. Run both doors end to end.

10. Grommets and leads

Fit the split grommets in the two low lead holes (160 above the deck) and run each original lead separately to the power point. No powerboard, nothing in a door track or under a machine foot.

11. Appliances in

Roll the fridge in, door hinge on the left, 107 back from the cabinet front. Stand the washer, fit the maker's stacking kit and lift the dryer on. Level all three on their own feet; the cabinet carries none of them.

Type	Item	Qty	Use
Sheet	Black film-faced formply 2400 x 1200 x 17, exterior (A-bond)	8 sheets	CNC supplier supplies and cuts (Cut list tab)
Sheet	Exterior edge sealer, black	1 L	Two coats on every cut edge and the door grooves
Sheet	PU sealant/adhesive, black (Sikaflex-11FC or similar)	4 x 300 ml	Housings, angles, hoods, pads
Hardware	Brio Wardrobe Double 50 kit WD50-15/2 (1500 double track, face-fix hangers, stops, 101N sawcut guides, screws)	1 kit, track cut to 1	Top-hung, 50 kg / 900 wide per panel, 16-35 thick; Brio stockists e.g. Keeler Hardware, Lincoln Sentry
Hardware	Aluminium angle 25 x 25 x 3, black anodised	4 x 1700	Inside the back corners
Hardware	Stainless angle 50 x 40 x 3	2 x 300	Hold-downs inside the cheeks
Hardware	Stainless insect mesh	1 m x 300	Behind the eight back openings
Timber	American walnut 32 x 20 x 400, from workshop offcuts	2	Door pulls
Timber	Exterior hardwood oil (e.g. Cutek or Intergrain UltraDeck Oil)	250 ml	Walnut pulls; seal end grain
Fixings	8g x 50 stainless countersunk timber screws	60	Through the top into the panel edges
Fixings	8g x 25 stainless countersunk	90	Corner and hold-down angles
Fixings	10g x 50 stainless countersunk	4	Hold-downs into the deck joists
Fixings	10g x 25 stainless countersunk	16	Pads
Fixings	6g x 16 stainless pan	32	Rain hoods
Fixings	8g x 30 stainless countersunk	6	Pulls, from behind
Fixings	Stainless staples, 10 mm	1 box	Mesh
Print	ASA filament, black	about 3.5 kg	Prints tab: 3 STLs, 23 prints

Checks, open items and ASA prints

	Check	Result
Pass	Fridge side clearance (maker 50)	72.5 each side
Pass	Fridge back clearance (maker 50)	186 to the back panel: the chimney behind the fridge
Pass	Fridge top clearance (maker 100)	183 to the top; the tracks sit in front of the fridge face
Pass	Stack under the top	238 above the dryer top
Pass	Laundry passes the open door	opening 841.5-1514, machines 859.0-1459.0
Pass	Rear pull clear of the closed front door	28 mm finger gap
Pass	Fridge door parks clear of the laundry pull	parked fridge door ends 8 short of the pull
Pass	Fridge opening with the door parked short	opening 17-709.5, fridge 89.5-684.5
Pass	Rear pull in the gap between the doors	pull face 6 behind the front line
Pass	Track behind the fascia	track y 20-72, z 1927-1968; fascia covers z 1928-1968
Pass	Brio panel limits (900 wide, 2400 high, 16-35 thick, 50	764.5 x 1915 x 17, about 16 kg
Pass	Rear track clear of the divider	28 mm
Pass	Leads exit near the power point	lead holes centred 160 above the deck; power point about 100
Pass	Fridge vents (Liebherr/Electrolux built-in rule: 200 cm2	per level per bay: openings 420 cm2 gross, about 294 through the mesh; hoo
Pass	Rain must climb to reach a vent	hood mouth faces down, 80 below the bottom of the opening
Pass	High vents above the appliances	high openings start 1820; fridge 1785, stack 1730
Pass	Doors clear of the deck	10 above the deck
Pass	Fridge door opens 115 deg past the cheek	closest 24 mm at 115 deg (simplified slab, left hinge)

Open item	Before the cut
Washer and dryer models	Provisional 600 x 680 x 850 and 600 x 670 x 850. Confirm both models and the maker's stacking kit; the dryer must be heat-pump or condenser.
Dryer air	Run the dryer with the laundry door slid open. Makers say a cupboard dryer needs louvred door openings or the door open while it runs (LG asks for two door openings of 387 cm2). The back vents help but are not a substitute.
Brio track geometry	Door lines, track set-out and the 46 mm fascia are modelled on Brio's published figures (panel height = opening - 53, 52 wide track, about 26 between door lines). Check the kit's data sheet B55 before releasing the fascia and door heights.
Hardware outdoors	Brio Wardrobe Double 50 is sold for wardrobes. Under cover it should be fine; spray the bearings with dry PTFE lube once a year and keep the track clear of leaves.
Power point	About 100 above the deck on the studio wall; the leads now leave through 60 mm holes 160 above the deck. Measure its position along the wall. An outdoor outlet must be weatherproof (IP-rated).
Rear door set-back	With bypass doors one door has to run behind the other: the laundry door sits 26 behind the flush fridge door.
Door flatness	1915 tall formply doors rely on their balanced film faces to stay flat. Keep both faces sealed; if one bows, a black aluminium angle screwed up its back edge fixes it.
Washer door	With the laundry door open, the parked door sits just left of the washer and 60 to 80 back from its face. A left-hinged washer door opened past about 120 degrees will touch it. Prefer a right-hinged washer, or open it to 90 degrees.

ASA print	Qty	Size	Notes
04_cable_grommet.stl	2	82 × 82 × 20	Split grommet for the two 60 mm lead holes (C2, C3). Smooth bore protects the leads. Print flange down. HOLD: confirm the plugs fit through 48 mm or enlarge the hole first.
05_edge_pad.stl	13	80 × 24 × 4	4 mm pad under the ply edges (13 of them). The box reads as flush to the deck but the ply never sits in water and the deck boards can dry. Bed in sealant, 1 x 10g x 25 into the deck. Print 100% infill.
06_rear_hood.stl	8	250 × 200 × 75	Rain hood over each back opening (8). Closed top, sides and face; open only at the bottom, 80 below the opening, so wind-driven rain has to climb to get in. Mouth 244 x 69 = 168 cm2. Staple stainless mesh behind the opening, bed the flange in sealant, 4 x 6g x 16. Print on its face (flange up), 4 walls; 250 wide: needs a 256 bed.